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Areas of physics by complexity



Newton's
Mechanics

Electro-
Magnetism

Special
Relativity

Quantum Mechanics
General Relativity

Quantum
Field Theory

Complexity
Science

Project # 40: Monthly snapshots of the Internet

AS-level network from BGP data

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The observed networks are snapshots of the internet taken as from one RIPE RIS (Routing Information Service) monthly from a period of time that goes from october 1999 to april 2026. The data does not comprehend all the endpoints, as it's a AS-level topology, where AS stands for Autonomous System and it is a massive, independent network controlled by a single administrative entity. The RIPE RIS project places passive servers (collectors) inside major internet exchange points. These collectors act like regular internet routers without forwarding any traffic. They simply listen to the BGP broadcasts sent to them by their neighbors and continuously log those AS_PATH strings into massive text files. Border Gateway Protocol (BGP) is a gateway protocol designed to exchange routing and reachability information among autonomous systems (AS) on the Internet. BGP is classified as a path-vector routing protocol, and it makes routing decisions based on paths, network policies, or rule-sets configured by a network administrator. A script automatically processes all of them, extracting every adjacent pair. If a pair is seen for the first time, it is added to the dataset as a new edge. If the pair has been seen before, the script simply adds +1 to its weight. By the end of the month, the script outputs the .tsv file. The list of unique pairs becomes an edge list, and the count of how many times each pair was observed in the AS_PATH becomes an edge weight.

The main analysis concerns the number of nodes and edges per month, detecting the growing trend and catching possible changes and missing data. The distribution of the size of the connected components will be then examined, detecting possible not fully connected networks. After confirming the heterogeneous structure of the network, the analysis will shift to a fit of the gathered degree data firstly with a powerlaw hypothesis, and then comparing qualitatively and with a loglikelihood ratio the results of the fit with other heavy-tail distribution proposals, assessing the evolution in time and the better performing models.

The examination will then shift to the appearing/disappearing/permanent node and edges, both looking at the amount of changed features over consecutive months and saving the specifics in data files. Particular attention will be given to the highest degree nodes. Mixture/Topological correlation over time will be then examined by means of the nearest neighbors degree distribution and it's scaling exponent. The review will then end with a modularity analisys, using the Leiden algorithm as baseline.

The graph of the trend of node and edges throughout the data collection period are in Fig. 1.1.

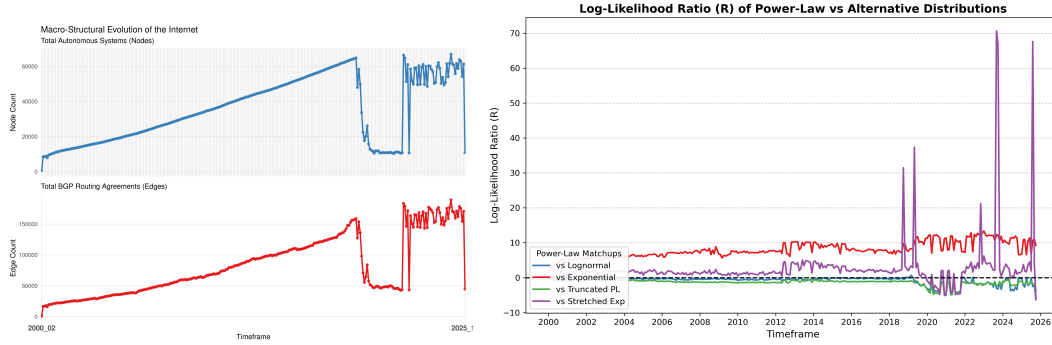


Figure 1.1: **On the left:** trend of nodes and edges count in time of the observed networks. The drop in the period between approximately the start of 2020 and 2022 shows a drop. As the routing data are taken from a single collector, the drop could be justifiable with the fact that one or more major providers decided to not show routing data for that period. **On the right:** log-likelihood ratio between heavy-tail distribution proposals (in green log-normal, in red exponential, in purple stretched exponential) and a power-law proposal for the degree distribution of the networks per month.

As for the connectedness of the network most of the network have the size of the largest connected component matching the total size of the network as expected. The few counter example (17 to be exact) could be due to the way edges are estimated in the data collection or due to an adding of a private connection to the data.

The degree distribution of all the networks present undoubtedly an heterogeneous structure. As a result the first hypothesis for a degree distribution is a power-law distribution. This would though mean that the network that models the internet would have a Robust Yet Fragile structure, meaning that targeted attacks would lead to a very disruptive behaviour (heuristic proofs of that in the Appendix). Given the data on degree distribution of the single networks, on each network are tested different heavy tailed distribution proposal (in particular a log-normal, a power-law with exponential cutoff, an exponential distribution and stretched exponential), using the *powerlaw* library, with an maximum likelihood approach and comparison made by log-likelihood ratio, results in Fig. 1.1. As it can be seen the power-law model with exponential cut-off and the lognormal model show on average better likelihood performance across the dataset, even though the most noticeable different is in the recent years (the 2020-2022 effect leads to more noisy results).

From the plots of the number of nodes/edges that persist/appear/disappear between

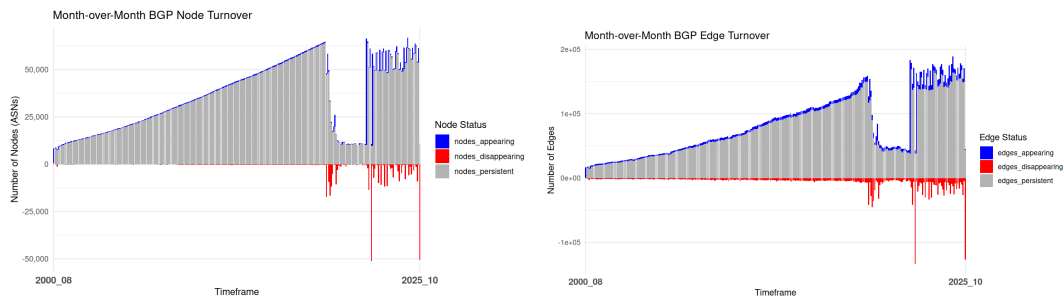


Figure 1.2: **On the left:** plot of consecutive nodes turn over per month, with in gray the number of persistent nodes, in blue the number of nodes that appeared and in red the number of disappearing nodes. **On the right:** plot of consecutive edges turn over per month, with in gray the number of persistent nodes, in blue the number of nodes that appeared and in red the number of disappearing nodes.

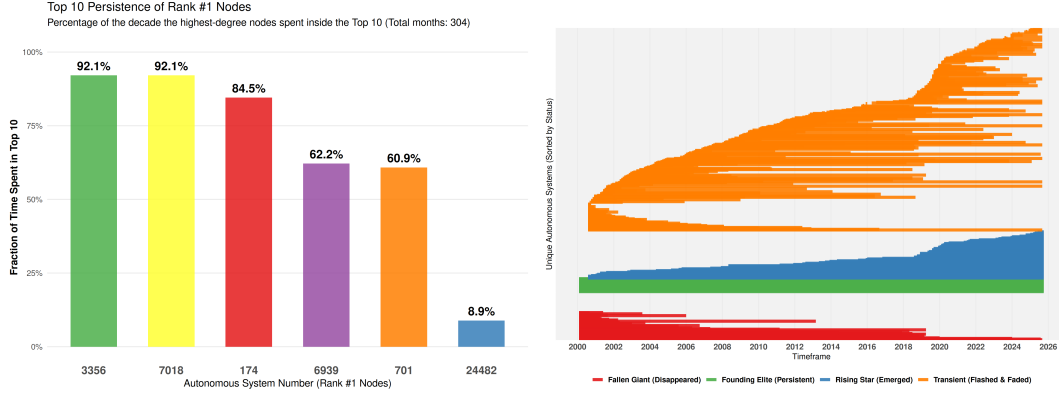


Figure 1.3: **On the left:** bar plot of the number of times over the all the time period the top nodes from each month remained in the top 10 nodes across all nodes. We can notice how the role of highest degree nodes is detained only by 6 nodes in the whole period, with on average an high persistence in the top 10 nodes with highest degree. **On the right:** appearance/disappearance and persistence of the top 100 nodes with highest degree per month and from each month over all the time frame.

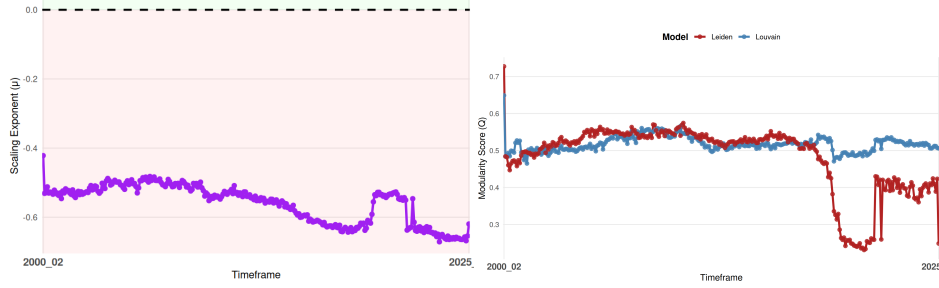


Figure 1.4: **On the left:** scaling of the nearest neighbor distribution for each month, with hypothesis $k^{(nn)}(k) \propto k^\mu$, obtained with a least squares linear regression in log-log scale. Negative values of μ indicates a disassortative mixture. **On the right:** modularity score comparison for each analyzed month for the found communities with the Louvain and Leiden algorithm. Although the fact that the Leiden algorithm seems to perform worse in different occasions, this is only due to the greedy moves made in the Louvain algorithms, that tends to create even weakly connected communities just to increase the modularity score.

each month in Fig. 1.2 it can be for sure said that the structure is stable, even though in recent years we can see that the variance rose undoubtedly. Although we can think that this could be due a change in routing procedures, it remains just an hypothesis.

The appearance and disappearance of the top 100/highest degree nodes in Fig. 1.3 per month suggests that the latter ones are few and persistent in the top 10 scoring nodes for most of the analyzed windows, as only 6 nodes cover the role of highest degree nodes over all the scanned monthly data.

From the data on assortativeness in each network in Fig. 1.4, computed with the analysis of the scaling of the nearest neighbor average degree as a function of the degree, we can firstly say that the networks are without a doubt disassortative. The data shows a growing disassortative characteristic during the window of time. The μ values are extracted with a least squares linear regression in log-log scale.

In order to identify modular structure in the networks it has been used the Leiden algorithm. The algorithm evaluates individual nodes and greedily moves them to neighboring communities to maximize immediate modularity (usually using Newman's modularity), but differently from the Louvain algorithm before forming the super-nodes the Leiden algorithm breaks them down in subcommunities to verify their connectiveness. The communities are then treated as super-nodes in a aggregated network. And the process continues iteratively. As a comparison the performance of the algorithm

in terms of modularity are put against the Louvain algorithm. The two models are then compared with a Stochastic Block Model, that identifies the community with a Minimum Description Length criterion.

2 | Appendix

- **Pre-processing:** the data pre-processing R code uses *rvest* for web scraping and *stringr*/*dplyr* libraries for text manipulation and *data.table* to handle efficiently the data, as built-in resources of R would slow down too much the computation. The code reads the hyperlinks present in the websites and finds the folder matching the requested dates and takes the "bview" files from the first day available file of the month. An awk program then reads the "AS_PATH" files and chop them into pairs, in a way that the smaller ASN is always printed first to avoid problem as the network is undirected. *bgpdump* is used to translate to text the binary files, then fed to the awk script. Then with *fread()* and *data.table* assign as weight to the unique edges the number of time they show up in the file. The sets present in the files are removed and the connection is simplifies by creating an edge between the node before the set and after it.
- **Community detection:** for this analysis it has been tested a Leiden and Louvain method for community detection both based on Newman's modularity to have an ulterior indicator of analysis. Alongside these, a Stochastic Block Model (SBM) was employed. Unlike Louvain and Leiden, which are algorithmic heuristics that seek densely connected assortative clusters, the SBM is a generative probabilistic model. Partitioning via SBM is achieved using statistical inference to find the block structure that minimizes the Minimum Description Length (MDL). Since comparing the models would be meaningless (as they are based on different criteria for modularity and even using the score the algorithm is based on can create counter-intuitive results as it can be seen with the modularity score report for the Louvain and Leiden algorithm) and without a source of ground truth the only possible analysis is a comparison of the pair-wise difference of the communities based on the Adjusted Rand Index (ARI). The results of the comparison are shown in Fig. 2.1.

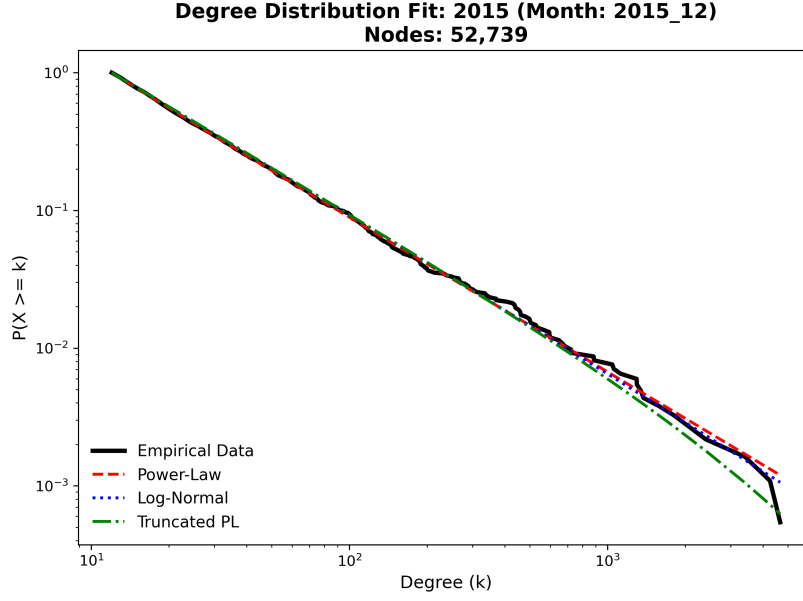


Figure 2.2: Fit of degree distribution in log scale with power-law, power-law with exponential decay, log-normal hypothesis.

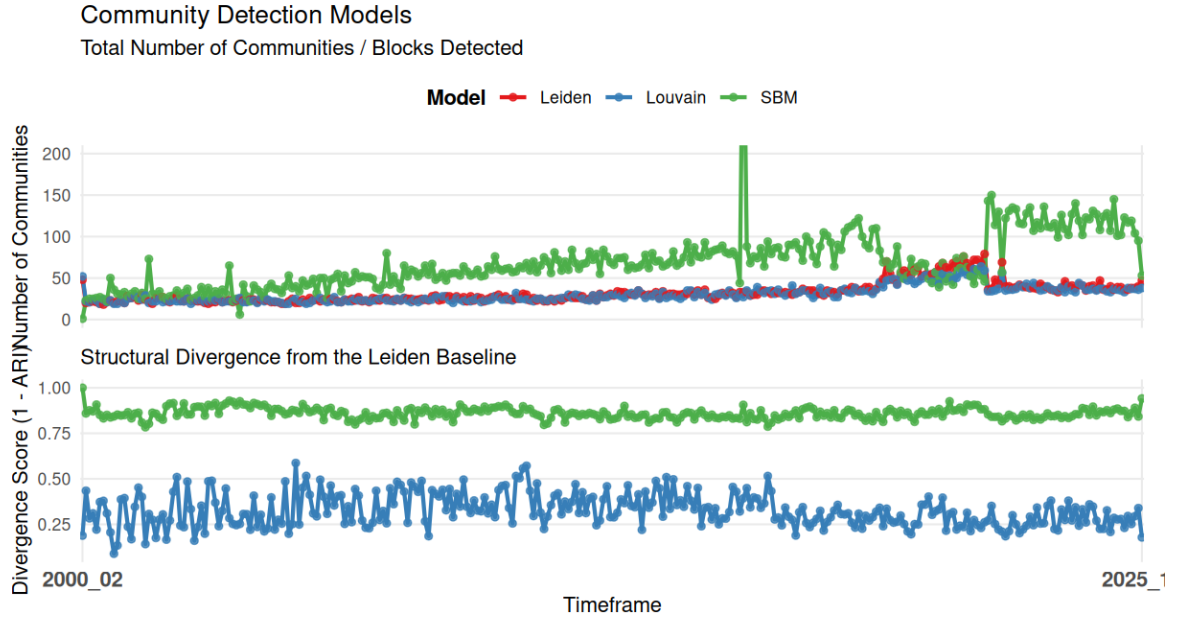


Figure 2.1: Comparison of different community detection algorithms, using a Louvain, Leiden and SBM model. On the top: the number of communities identified by each model. It can be seen how on average the SBM detects more communities, while the Louvain and Leiden algorithm detects about the same amount. On the bottom: Adjust Rand Index of the SBM and Louvain algorithm, using as a baseline the Leiden algorithm results. As this is a measure of similarity, it is expected that the Louvain method performs very similarly with respect to the Leiden method, although the difference between the two algorithms is noticeable.

- **Heterogeneous structure:** As a proof of the claims about the heterogeneous structure and compatibility with the proposed distribution an example is presented in Fig. 2.2.

- **Robust-Yet-Fragile behavior:** to address the fact that the possible distribution that could model the degree distribution of the networks have a Robust-Yet-Fragile behavior we show the behavior of the models in term of connectiveness (as size of the LCC) as a function of the occupation probability ϕ against an ER model with same average degree. To show this behavior the percolation is performed in a random failure scenario and in a selective attack, by removing in the second case nodes starting from the one with highest degree. The analysis uses the interpolation data for the different models and simulates multiple synthetic data following that model, averaging values to remove randomness. The interpolation data is taken from three separated time instances (also the interval 2020-2022 is avoided). The results are in Fig. 2.3.

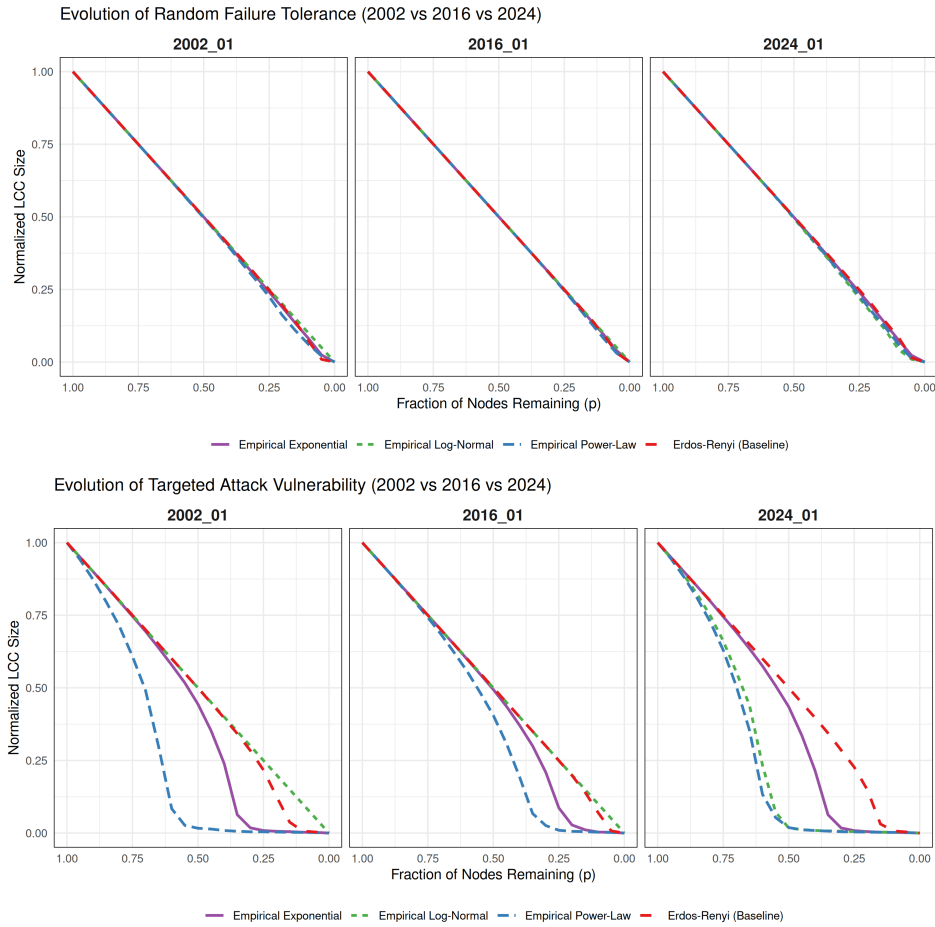


Figure 2.3: simulation of percolation for a log-normal, power-law and truncated power-law model against an ER model with the same average degree. The above plot shows the case of random failures, while the plot below shows the case of targeted attacks, removing firstly the highest degree nodes.